

Name: _____

Class: _____



Mr. Rawson • GCSE Computer Science

Year 10 2026

The Byte Strip

Eight cards. No pens, no arithmetic, no subtraction.

Start here. Cut the eight cards off the bottom of this sheet. Lay them on the mat below, **biggest on the left**, one card per square.

The back of every card is blank, and that is the point. A card **face up** means that place value is switched **on** — a 1. A card turned **blank side up** means **off** — a 0. The square underneath tells you which card it is, so you never lose your place.

Move 1 Show me a number. (denary → binary)

Start with every card blank side up. Mr. Rawson calls a number. Turn over the **biggest card that fits**. Then the next biggest that fits. Keep going until the cards showing add up to exactly his number. **Never turn a card that would take you over.** Read the row left to right — that is the binary.

Move 2 Read me a number. (binary → denary)

Mr. Rawson calls out eight digits. Set the cards to match, then **add up what is face up**. Same cards, run backwards.

Move 3 Split the strip. Slide the four right-hand cards away from the four left-hand cards. **Each group of four is a nibble.** Go to page 2.

Move 4 Build the strip you keep. Fill in the page 2 table with your own cards. **Keep it** — it is your reference for the rest of this unit, for the **quiz on Thu 17 Sept** and for the **test on Tue 6 Oct**.

THE BYTE MAT • one card per square

128	64	32	16	8	4	2	1

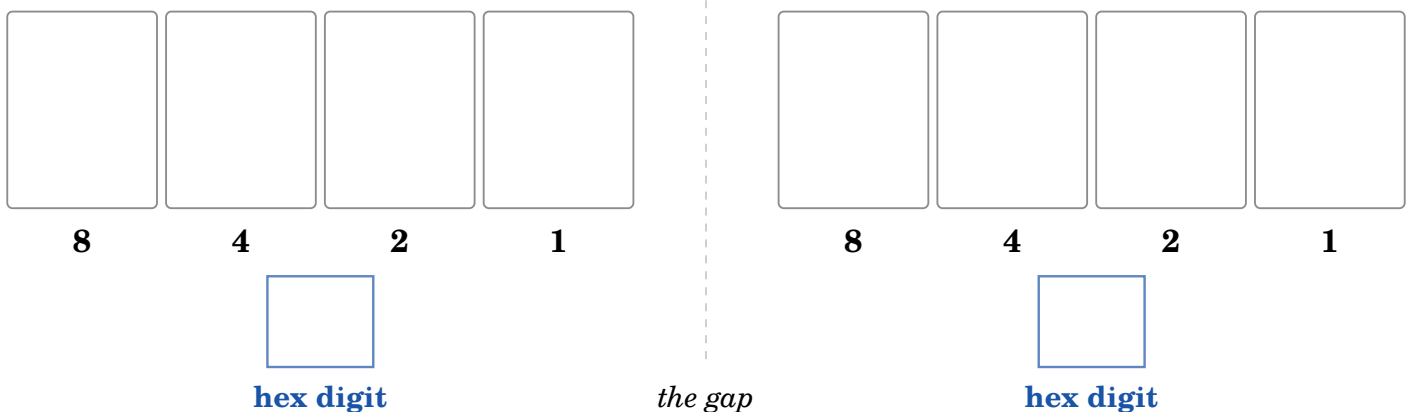
✂ Cut along the dashed line, then cut the eight cards apart.

128	64	32	16	8	4	2	1
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Move 3 · Split the strip

Name: _____

Put four cards on each mat below and **read each mat on its own**. The left-hand four were 128, 64, 32, 16 on the byte mat — on their own mat they read 8 4 2 1, exactly like the right-hand four. **A nibble is a nibble wherever it sits**, and each one can only show **0 to 15**.



The one number that shows the whole idea

$\begin{array}{c} 0011 \\ \underbrace{\hspace{1cm}} \\ 3 \end{array}$
 $\begin{array}{c} 0100 \\ \underbrace{\hspace{1cm}} \\ 4 \end{array}$

Read it as **two nibbles**: $0011 = 3$, $0100 = 4 \rightarrow 34$ in hexadecimal

Read it as **one byte**: $32 + 16 + 4 \rightarrow 52$ in denary

These are not two different numbers. One pattern of eight cards, written down two ways: **34** in hex is **52** in denary. Check it: $3 \times 16 + 4 = 52$.

Hexadecimal is just where you put the gap. Nothing about the cards changed.

Your turn. Set your cards to 1101 1110.

a) In hexadecimal it is _____ b) In denary it is _____ c) Check: $___ \times 16 + ___ =$ _____

Move 4 · Your reference strip — build it, keep it

Use **one** nibble mat. Make every number from 0 to 15 with your four cards, and write down what you see. Two rows are done for you.

Denary	8	4	2	1	Hex
0	0	0	0	0	0
1					
2					
3					
4					
5					
6					
7					

Denary	8	4	2	1	Hex
8					
9					
10	1	0	1	0	A
11					
12					
13					
14					
15					

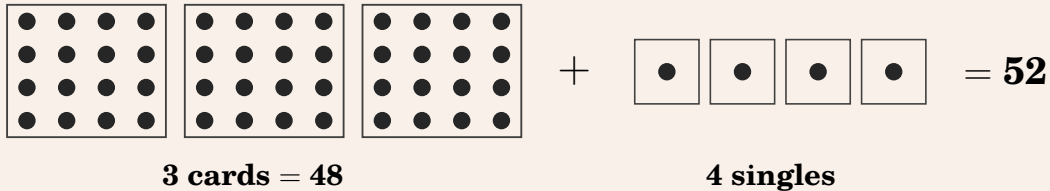
Four cards run out at 15 — which is the whole reason hexadecimal needs letters. Sixteen values, only ten digits. **Now turn over — what is a nibble actually worth?**

Move 5 • The sixteen-card

On page 2 you split 0011 0100 into two nibbles and read **3** and **4**. But the left nibble was never just **3**. It was **3 sixteens**. Turn to the last page, cut out the cards, and come back.

In a hex pair, the left digit counts *cards* and the right digit counts *dots*.

34 — a number you have already met



Same number, third time. On page 2 this byte was 0011 0100, and $32 + 16 + 4 = 52$. In hex it is 34. On the desk it is three cards and four singles. **Nothing has changed but how you are looking at it.**

Your turn. Build each one, then check the arithmetic.

	Cards $\times 16$	Singles	Total in denary
4B	_____	_____	_____
7C	_____	_____	_____

Count out sixteen singles.

Now put them beside **one card**. They are worth exactly the same. **Sixteen singles are one card** — so swap them, and the right-hand digit goes back to nought while the left-hand one goes up by one.

That swap is the only thing place value has ever meant.

DE will not fit, and that is deliberate.

D is **13** — more cards than you have. **So stop building and start calculating:** $13 \times 16 = 208$, plus **14**, is **222**.

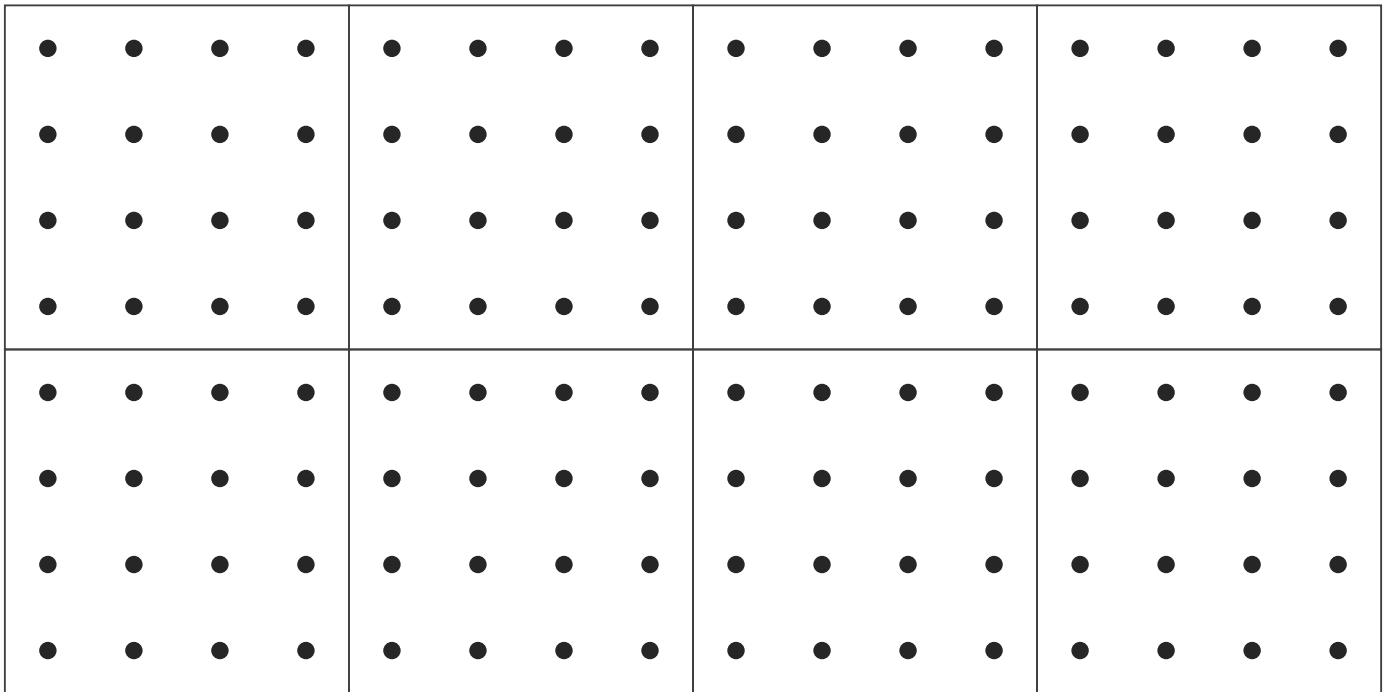
Check it against page 2. The cards were never the point; they were how you learn to trust the arithmetic.

With the pair next to you, put all sixteen cards on the desk. Eight and eight. That is $16 \times 16 = 256$ dots — but the biggest number a byte can hold is **255**, because FF is **15 cards and 15 singles** = $240 + 15 = 255$. **So the sixteenth card is one too many.** A byte counts **0 to 255**: 256 different values, and the two of you have just laid every one of them on the table. **And next: a colour code is three byte strips side by side — each channel gets its own sixteen cards.**

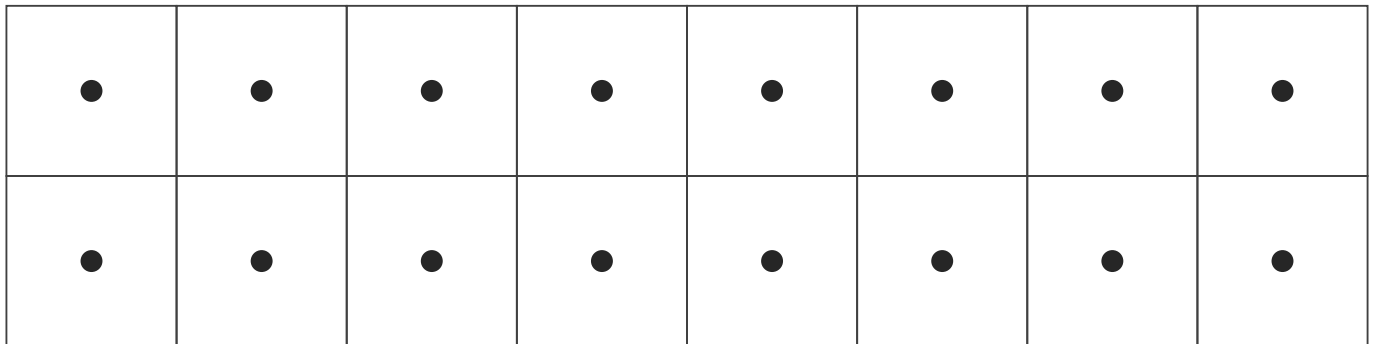
Cut these out

Cut the two blocks apart first, then cut each one up. Eight cards, sixteen singles.

✂ **THE SIXTEEN-CARDS** — 16 dots on each. Three cuts down, one across the middle.



✂ **THE SINGLES** — one dot each, half the width of a card. Seven cuts across, one down the middle.



Eight cards is 128 dots, and sixteen singles is sixteen more. Put your set beside another pair's and you have every value a single byte can hold.